Math 105L In Class 1 Practice Solutions

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Crash Course in Trig P1 and P2

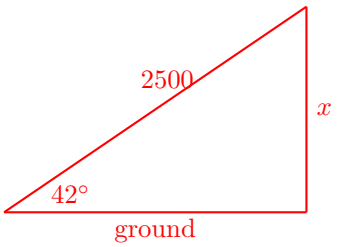
For each of questions 1-10, draw a picture first, label it well, then compute. The pictures will be part of the grading on the test!

1. A drone takes off from level ground and rises at a constant angle of 42°. After traveling 2,500 feet along its path, how high above the ground is the drone?

   Let x be the height of the drone.

   $$\sin 42^\circ = \frac{x}{2500}$$

   $$x = 2500 \sin 42^\circ \approx 1673 \text{ ft}$$

2. A wheelchair ramp is built at an angle of 6° with the horizontal. If the platform is 3 feet above ground level, how long is the ramp?

   Let L be the length of the ramp.

   $$\sin 6^\circ = \frac{3}{L}$$

   $$L = \frac{3}{\sin 6^\circ} \approx 28.7 \text{ ft}$$

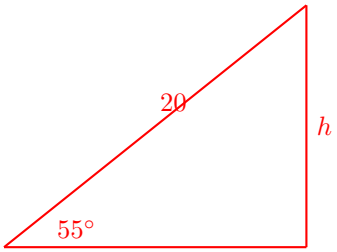
   

3. A 20-foot ladder leans against a building, making a 55° angle with the ground. How high up the wall does the ladder reach?

   Let h be the height.

   $$\sin 55^\circ = \frac{h}{20}$$

   $$h = 20 \sin 55^\circ \approx 16.4 \text{ ft}$$

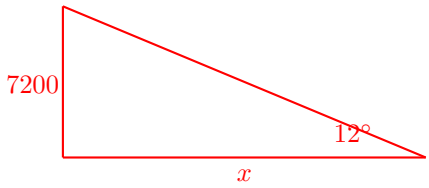
   

4. An airplane descends at an angle of 12°. If its altitude is 7,200 feet, how far horizontally is it from the airport?

Let x be the horizontal distance.

$$\tan 12^\circ = \frac{7200}{x}$$

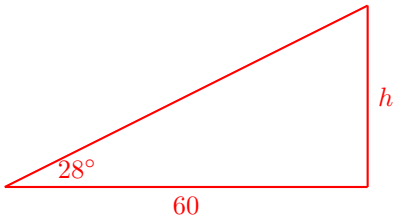
$$x = \frac{7200}{\tan 12^\circ} \approx 33{,}900 \text{ ft}$$



5. A surveyor stands 60 meters from the base of a cliff. The angle of elevation is 28°. Find the height of the cliff.

$$\tan 28^\circ = \frac{h}{60}$$

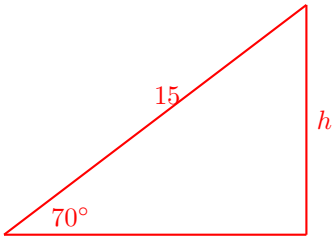
$$h = 60 \tan 28^\circ \approx 31.9 \text{ m}$$



6. A rope 15 meters long is tied to the top of a pole. The rope makes a 70° angle with the ground. How tall is the pole?

$$\sin 70^\circ = \frac{h}{15}$$

$$h = 15 \sin 70^\circ \approx 14.1 \text{ m}$$



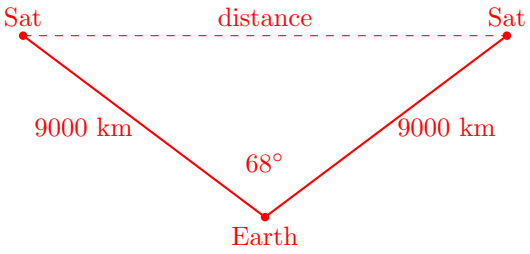
7. Two satellites are each 9,000 km from Earth and subtend an angle of 68°. Estimate the distance between them.

Note: 'subtend an angle' means the angle between the lines of sight from Earth to the satellites here. There was a similar problem on the worksheet!

Each half-angle is 34°.

$$\sin 34^\circ = \frac{x}{9000}$$

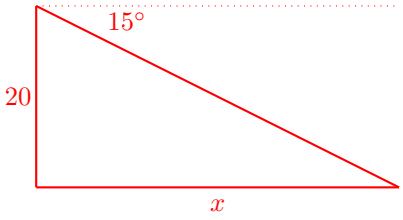
$$\text{distance} = 2x = 2(9000 \sin 34^\circ) \approx 10{,}070 \text{ km}$$



8. From the top of a lighthouse 20 meters above sea level, the angle of depression to a boat is 15°. Find the distance to the boat.

$$\tan 15^\circ = \frac{20}{x}$$

$$x = \frac{20}{\tan 15^\circ} \approx 74.6 \text{ m}$$



9. Angles of elevation to the top of a tower are 38° and 22° from points 25 meters apart. Find the height of the tower.

Let d be the distance from the closer point.

$$\tan 38^\circ = \frac{h}{d}, \quad \tan 22^\circ = \frac{h}{d+25}$$

So $h = d\tan 38^\circ$, giving

$$\tan 22^\circ = \frac{d\tan 38^\circ}{d+25}.$$

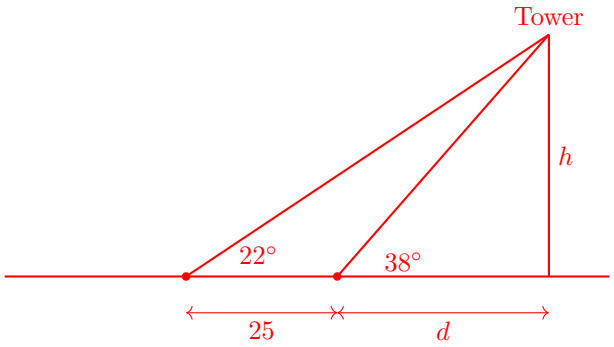
$d\tan 22^\circ + 25\tan 22^\circ = d\tan 38^\circ$, or

$$25\tan 22^\circ = d(\tan 38^\circ - \tan 22^\circ).$$

This gives:

$$d = \frac{25\tan 22^\circ}{\tan 38^\circ - \tan 22^\circ} \approx 26.8 \text{ m}$$

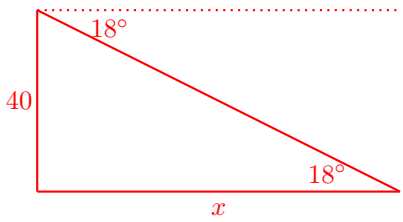
Finally, $h = d\tan 38^\circ \approx 20.9$ m.



10. From a cliff 40 meters high, the angle of depression to a buoy is 18°. How far is the buoy from the base?

$$\tan 18^\circ = \frac{40}{x}$$

$$x = \frac{40}{\tan 18^\circ} \approx 123 \text{ m}$$



11. Let the angle between the positive x-axis and the line $y = -10x$ be θ. Find the exact value of $\tan\theta$. What are all possible pairs of values of $\sin\theta$ and $\cos\theta$?

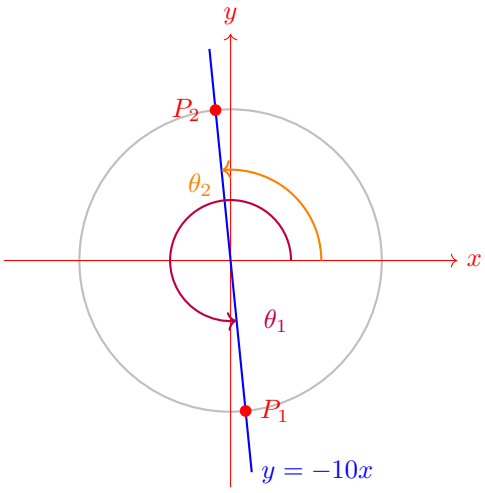
The tangent of an angle θ in standard position is defined by the slope of its terminal side. For the line $y = -10x$:

$$\tan\theta = -10$$

The terminal side of θ intersects the unit circle $x^2+y^2 = 1$ at points where the coordinates $(\cos\theta, \sin\theta)$ must satisfy the line's equation. Substituting $y = -10x$ into the unit circle equation:

$$x^2 + (-10x)^2 = 1 \implies 101x^2 = 1 \implies x = \pm\frac{1}{\sqrt{101}}$$

This gives two possible intersection points on the unit circle, corresponding to the two possible terminal sides of the line:



Since $(\cos\theta, \sin\theta)$ are the coordinates of the intersection of the terminal side and the unit circle, we have two possible pairs:

(a) **Quadrant IV (θ_1):** $x > 0$ and $y < 0$.

$$\cos\theta = \frac{1}{\sqrt{101}}, \quad \sin\theta = -\frac{10}{\sqrt{101}}$$

(b) **Quadrant II (θ_2):** $x < 0$ and $y > 0$.

$$\cos\theta = -\frac{1}{\sqrt{101}}, \quad \sin\theta = \frac{10}{\sqrt{101}}$$

Final Values:

$$\tan\theta = -10$$

Possible pairs $(\sin\theta, \cos\theta)$:

$$\left(-\frac{10}{\sqrt{101}}, \frac{1}{\sqrt{101}}\right) \text{ and } \left(\frac{10}{\sqrt{101}}, -\frac{1}{\sqrt{101}}\right)$$

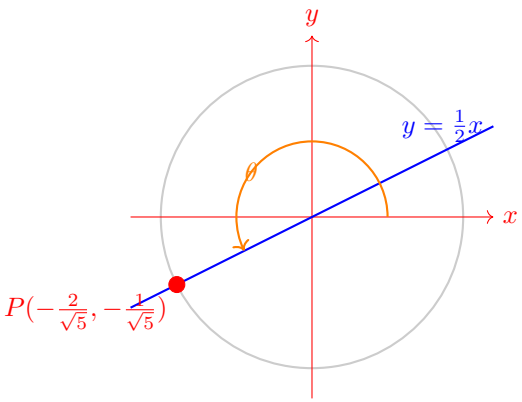
12. Let the angle between the positive x-axis and the line $y = \frac{1}{2}x$ be θ. If the terminal side of θ lies in the third quadrant, find the exact values of $\csc\theta$, $\sec\theta$, and $\cot\theta$.

The line $y = \frac{1}{2}x$ has a slope $m = \frac{1}{2}$, which implies $\tan\theta = \frac{1}{2}$. To find the coordinates $(\cos\theta, \sin\theta)$, we find the intersection of the line and the unit circle $x^2 + y^2 = 1$:

$$x^2 + \left(\frac{1}{2}x\right)^2 = 1 \implies \frac{5}{4}x^2 = 1 \implies x^2 = \frac{4}{5}$$

Since θ is in the **Third Quadrant**, both x and y must be negative:

$$x = \cos\theta = -\frac{2}{\sqrt{5}}, \quad y = \sin\theta = -\frac{1}{\sqrt{5}}$$



Using the reciprocal identities:

- $\sec\theta = \frac{1}{\cos\theta} = -\frac{\sqrt{5}}{2}$
- $\csc\theta = \frac{1}{\sin\theta} = -\sqrt{5}$
- $\cot\theta = \frac{1}{\tan\theta} = 2$

13. A line makes an angle of θ with the positive x-axis. Suppose $\sec\theta = -\frac{5}{3}$ and $\sin\theta < 0$. Find the equation of the line and calculate the value of $\cot\theta$.

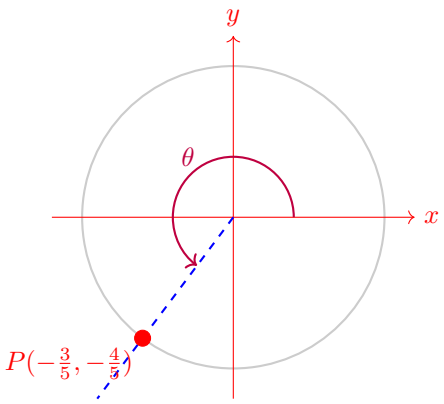
First, find $\cos\theta$ using the reciprocal relationship:

$$\cos\theta = \frac{1}{\sec\theta} = -\frac{3}{5}$$

Using the Pythagorean identity $\cos^2\theta + \sin^2\theta = 1$:

$$\left(-\frac{3}{5}\right)^2 + \sin^2\theta = 1 \implies \sin^2\theta = \frac{16}{25}$$

Given $\sin\theta < 0$, we have $\sin\theta = -\frac{4}{5}$.



The slope of the terminal side is:

$$m = \frac{\sin\theta}{\cos\theta} = \frac{-4/5}{-3/5} = \frac{4}{3}$$

The equation of the line is $y = \frac{4}{3}x$. Finally:

$$\cot\theta = \frac{1}{\tan\theta} = \frac{3}{4}$$

14. A geostationary orbit is one of altitude (distance above the surface of the earth) 22,236 miles. Given that the radius of the Earth is 3,959 miles, find the radian measure of an arc of length 500 miles along its orbit. Write your answer to two significant figures.

To find the radian measure θ of an arc, we use the formula for arc length s:

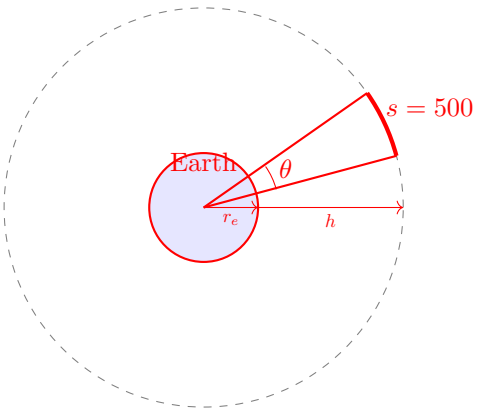
$$s = r\theta \implies \theta = \frac{s}{r}$$

where s is the arc length and r is the total radius of the orbit.

The radius of the orbit (r_{orbit}) is the sum of the Earth's radius (r_{earth}) and the altitude of the satellite (h):

$$r_{orbit} = r_{earth} + h$$

$$r_{orbit} = 3,959 + 22,236 = 26,195 \text{ miles}$$



Now, substitute the arc length $s = 500$ and the orbital radius $r = 26,195$ into the formula:

$$\theta = \frac{500}{26,195}$$

Simplifying the fraction:

$$\theta \approx 0.0190876 \text{ radians}$$

To the nearest thousandth:

$$\theta \approx 0.019 \text{ radians}$$

15. A circular race track has a radius of $1,200$ feet. A car starts at the point $(1200, 0)$ and travels counter-clockwise along the track for an arc length of $1,600$ feet.

    (a) Find the radian measure of the angle θ, from the horizontal to the car's final position. Then convert your answer to degrees to one decimal place.

    (b) Find the coordinates (x, y) of the car's position to two decimal places.

a. Radian Measure

The relationship between the arc length (s), the radius (r), and the angle in radians (θ) is defined by the formula:

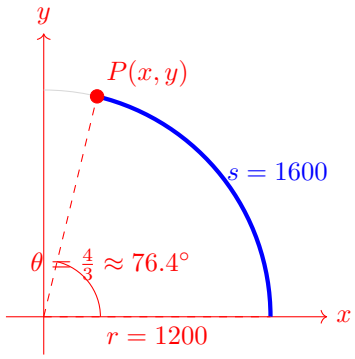
$$s = r\theta$$

Substituting the given values $s = 1600$ and $r = 1200$:

$$1600 = 1200\theta$$

$$\theta = \frac{1600}{1200} = \frac{4}{3} \text{ radians}$$

This is $\frac{4}{3} \times \frac{180}{\pi} = \frac{240}{\pi} \approx 76.4°$



b. Coordinates

On the unit circle, the coordinates are given by

$$x = \cos\theta \quad \text{and} \quad y = \sin\theta$$

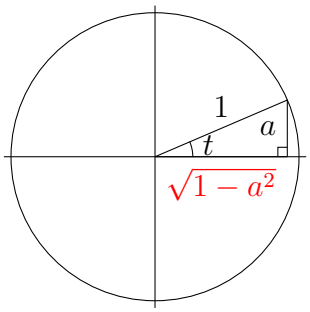
Therefore, on a circle of radius 1,200 ft, we get:

$$x = 1200\cos\left(\frac{4}{3}\right)$$

$$y = 1200\sin\left(\frac{4}{3}\right)$$

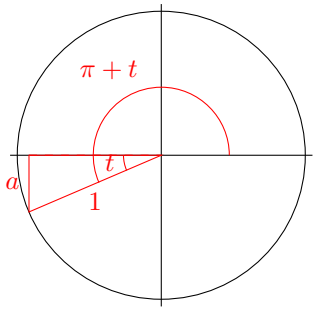
The exact coordinates of the car's position are $\left(1200\cos\left(\frac{4}{3}\right), 1200\sin\left(\frac{4}{3}\right)\right) \approx (282.28, 1166.32)$.

16. Let t be an angle in the first quadrant, as pictured below. For each of these, draw the given angle and a corresponding triangle congruent to the one given. Then use that to evaluate the given expression in terms of a.

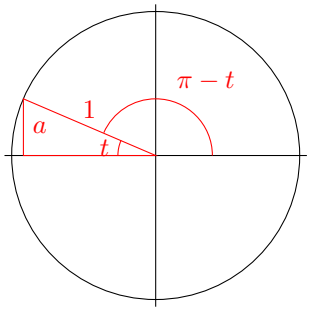


Note that the third length in the triangle is $\sqrt{1-a^2}$, by Pythagoras.

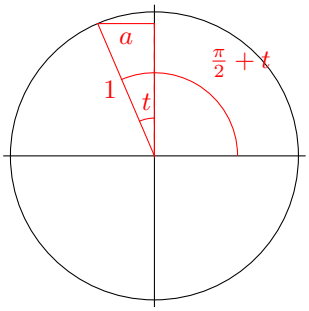
(a) $\cos(\pi + t) =$ $-\sqrt{1-a^2}$



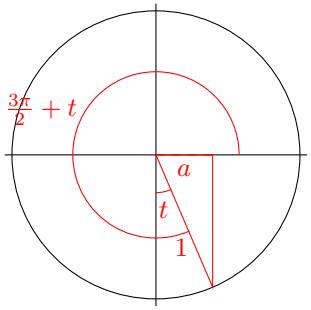
(c) $\tan(\pi - t) =$ $-\frac{a}{\sqrt{1-a^2}}$



(b) $\sin\left(\frac{\pi}{2} + t\right) =$ $\sqrt{1-a^2}$



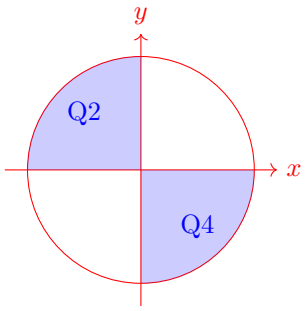
(d) $\sec\left(\frac{3\pi}{2} + t\right) =$ $\frac{1}{a}$



17. For what values of θ between $0°$ and $360°$ are each of the following true? Explain your answers with words and/or drawings.

(a) $\tan(\theta) < 0$

Since $\tan\theta = \frac{y}{x}$, the tangent is negative when x and y have opposite signs. This occurs in Quadrant II ($x < 0, y > 0$) and Quadrant IV ($x > 0, y < 0$).



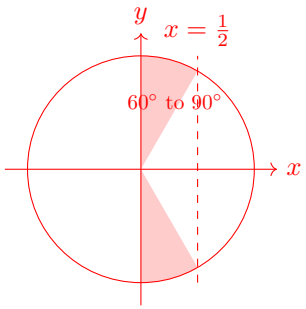
Result: $90° < \theta < 180°$ and $270° < \theta < 360°$.

(b) $\sec(\theta) \geq 2$

$\sec\theta = \frac{1}{\cos\theta}$. The inequality $\sec\theta \geq 2$ implies $\frac{1}{\cos\theta} \geq 2$.

Since we need the result to be positive, $\cos\theta$ must be positive, so $0 < \cos\theta \leq \frac{1}{2}$.

This occurs in Q1 and Q4 where the x-coordinate is between 0 and 0.5. $\cos\theta = 0$ when $\theta = 90°$ or $\theta = 180°$, and $\cos\theta = \frac{1}{2}$ when $\theta = 60°$ or $\theta = 300°$.



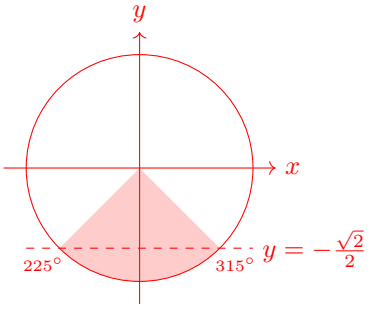
Result: $60° \leq \theta < 90°$ and $270° < \theta \leq 300°$.

(c) $\sin(\theta) \leq -\frac{\sqrt{2}}{2}$

$\sin\theta$ is the y-coordinate. We want the points on the unit circle where $y \leq -\frac{\sqrt{2}}{2}$.

$\sin\theta = -\frac{\sqrt{2}}{2}$ when $\theta = 225°$ and $\theta = 315°$.

So $\sin(\theta) \leq -\frac{\sqrt{2}}{2}$ starting from 225° and ending at 315°.



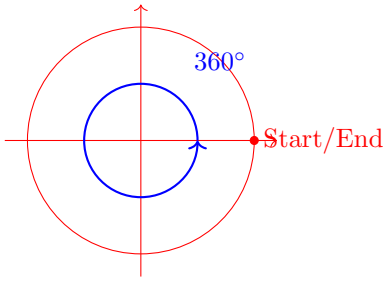
Result: $225° \leq \theta \leq 315°$.

Trig Identities and Limits

1. For each of the following, determine the smallest value $P > 0$ such that the statement is true for all θ in the domain. Explain your reasoning using the unit circle or the definition of the functions.

   (a) $\sin(\theta + P) = \sin(\theta)$ and $\cos(\theta + P) = \cos(\theta)$

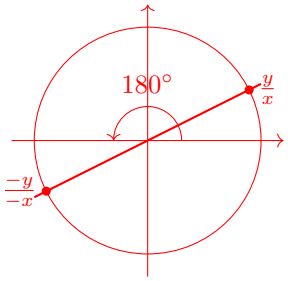
   Sine and Cosine represent the y and x coordinates on the unit circle. To return to the exact same point (x, y) and repeat the values, you must complete one full revolution around the circle.



   Result: $P = 360°$ (or 2π radians). This same period applies to their reciprocals, **cosecant** (csc) and **secant** (sec).

   (b) $\tan(\theta + P) = \tan(\theta)$

   Tangent is the ratio $\frac{y}{x}$ (the slope of the terminal side). A line has the same slope when it points in the opposite direction (a 180° rotation). For example, in Q1 both x, y are positive, and in Q3 both are negative, resulting in the same positive ratio.



   Result: $P = 180°$ (or π radians). This same period applies to its reciprocal, **cotangent** (cot).

   (c) Summarize the periods of all six trigonometric functions.

   Based on the symmetry of the unit circle:

Function	Period (Degrees)	Period (Radians)
$\sin\theta, \cos\theta$	360°	2π
$\csc\theta, \sec\theta$	360°	2π
$\tan\theta, \cot\theta$	180°	π

   Tangent and Cotangent repeat more frequently because the ratio of coordinates repeats every half-turn, whereas the individual coordinates themselves require a full turn to return to their original signs and values.

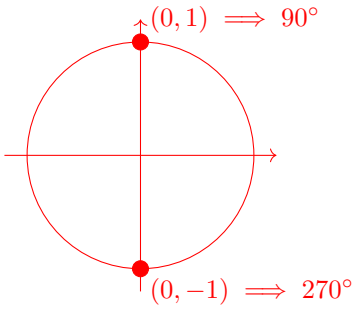
2. For each of the six trigonometric functions, determine the locations of their vertical asymptotes (VAs) between 0° and 360°. Explain your reasoning by identifying which coordinate (x or y) in the denominator causes the function to be undefined.

(a) Which functions have no vertical asymptotes?

$\sin\theta$ and $\cos\theta$ have no vertical asymptotes. On the unit circle, these are defined as the y and x coordinates respectively (with a radius of 1). Since the denominator is never zero, these functions are continuous for all real numbers.

(b) At what values of θ do $\tan\theta$ and $\sec\theta$ have vertical asymptotes?

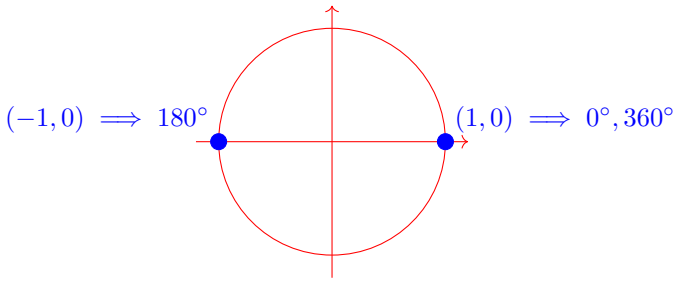
Both $\tan\theta = \frac{y}{x}$ and $\sec\theta = \frac{1}{x}$ have x in the denominator. Therefore, VAs occur whenever the x-coordinate on the unit circle is 0. This happens at the "top" and "bottom" of the circle.



Result: VAs at $\theta = 90°$ and $\theta = 270°$.

(c) At what values of θ do $\cot\theta$ and $\csc\theta$ have vertical asymptotes?

Both $\cot\theta = \frac{x}{y}$ and $\csc\theta = \frac{1}{y}$ have y in the denominator. Therefore, VAs occur whenever the y-coordinate on the unit circle is 0. This happens at the "left" and "right" sides of the circle.



Result: VAs at $\theta = 0°, 180°$, and $360°$.

3. Use the sum and difference formulas below to find the exact values of the given expressions.

$$\begin{aligned}
\sin(A+B) &= \sin A\cos B + \cos A\sin B \\
\sin(A-B) &= \sin A\cos B - \cos A\sin B \\
\cos(A+B) &= \cos A\cos B - \sin A\sin B \\
\cos(A-B) &= \cos A\cos B + \sin A\sin B
\end{aligned}$$

(a) Find the exact value of $\sin(15°)$.

We can represent $15°$ as the difference $A - B$, where $A = 45°$ and $B = 30°$.

$$\begin{aligned}
\sin(15°) &= \sin(45° - 30°) \\
&= \sin 45°\cos 30° - \cos 45°\sin 30° \\
&= \left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{3}}{2}\right) - \left(\frac{\sqrt{2}}{2}\right)\left(\frac{1}{2}\right) \\
&= \frac{\sqrt{6}}{4} - \frac{\sqrt{2}}{4} = \frac{\sqrt{6}-\sqrt{2}}{4}
\end{aligned}$$

(b) Find the exact value of $\cos(15°)$.

Using $A = 45°$ and $B = 30°$ in the cosine difference formula:

$$\begin{aligned}\cos(15°) &= \cos(45° - 30°)\\ &= \cos 45° \cos 30° + \sin 45° \sin 30°\\ &= \left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{3}}{2}\right) + \left(\frac{\sqrt{2}}{2}\right)\left(\frac{1}{2}\right)\\ &= \frac{\sqrt{6}+\sqrt{2}}{4}\end{aligned}$$

(c) Find the exact value of $\sin(75°)$.

We can represent $75°$ as the sum $A + B$, where $A = 45°$ and $B = 30°$.

$$\begin{aligned}\sin(75°) &= \sin(45° + 30°)\\ &= \sin 45° \cos 30° + \cos 45° \sin 30°\\ &= \left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{3}}{2}\right) + \left(\frac{\sqrt{2}}{2}\right)\left(\frac{1}{2}\right)\\ &= \frac{\sqrt{6}+\sqrt{2}}{4}\end{aligned}$$

(d) Find the exact value of $\cos(75°)$.

Using $A = 45°$ and $B = 30°$ in the cosine sum formula:

$$\begin{aligned}\cos(75°) &= \cos(45° + 30°)\\ &= \cos 45° \cos 30° - \sin 45° \sin 30°\\ &= \left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{3}}{2}\right) - \left(\frac{\sqrt{2}}{2}\right)\left(\frac{1}{2}\right)\\ &= \frac{\sqrt{6}-\sqrt{2}}{4}\end{aligned}$$

(e) Using your answers for $\sin(15°)$ and $\cos(15°)$, find the exact value of $\tan(15°)$.

From the previous problems, we know:

$$\sin(15°) = \frac{\sqrt{6}-\sqrt{2}}{4} \quad \text{and} \quad \cos(15°) = \frac{\sqrt{6}+\sqrt{2}}{4}$$

Substituting these into the tangent ratio:

$$\tan(15°) = \frac{\frac{\sqrt{6}-\sqrt{2}}{4}}{\frac{\sqrt{6}+\sqrt{2}}{4}} = \frac{\sqrt{6}-\sqrt{2}}{\sqrt{6}+\sqrt{2}}$$

To simplify, we multiply the numerator and denominator by the conjugate $(\sqrt{6} - \sqrt{2})$:

$$\tan(15°) = \frac{(\sqrt{6}-\sqrt{2})^2}{(\sqrt{6}+\sqrt{2})(\sqrt{6}-\sqrt{2})} = \frac{6 - 2\sqrt{12} + 2}{6-2} = \frac{8-4\sqrt{3}}{4} = 2 - \sqrt{3}$$

(Note: doing the rationalization isn't required here – I just included it to make things look nicer!)

(f) Using your answers for $\sin(75°)$ and $\cos(75°)$, find the exact value of $\tan(75°)$.

From the previous problems, we know:

$$\sin(75°) = \frac{\sqrt{6}+\sqrt{2}}{4} \quad \text{and} \quad \cos(75°) = \frac{\sqrt{6}-\sqrt{2}}{4}$$

Substituting these into the tangent ratio:

$$\tan(75°) = \frac{\frac{\sqrt{6}+\sqrt{2}}{4}}{\frac{\sqrt{6}-\sqrt{2}}{4}} = \frac{\sqrt{6}+\sqrt{2}}{\sqrt{6}-\sqrt{2}}$$

Rationalizing the denominator using the conjugate $(\sqrt{6}+\sqrt{2})$:

$$\tan(75°) = \frac{(\sqrt{6}+\sqrt{2})^2}{(\sqrt{6}-\sqrt{2})(\sqrt{6}+\sqrt{2})} = \frac{6+2\sqrt{12}+2}{6-2} = \frac{8+4\sqrt{3}}{4} = 2+\sqrt{3}$$

(Note: doing the rationalization isn't required here – I just included it to make things look nicer!)

4. Using the facts that $\sin(2A) = 2\sin A\cos A$ and $\cos(2A) = \cos^2 A - \sin^2 A$, and the angle addition formulas above, find formulas for the following in terms of $\sin A$ and/or $\cos A$:

(a) $\sin(3A)$

We can rewrite $3A$ as $(2A + A)$. Applying the sine sum formula:

$$\begin{aligned}\sin(3A) &= \sin(2A+A)\\ &= \sin(2A)\cos A + \cos(2A)\sin A\end{aligned}$$

Now substitute the double angle identities $\sin(2A) = 2\sin A\cos A$ and $\cos(2A) = \cos^2 A - \sin^2 A$:

$$\begin{aligned}\sin(3A) &= (2\sin A\cos A)\cos A + (\cos^2 A - \sin^2 A)\sin A\\ &= 2\sin A\cos^2 A + \sin A\cos^2 A - \sin^3 A\\ &= 3\sin A\cos^2 A - \sin^3 A\end{aligned}$$

(b) $\sin(4A)$

We can treat $4A$ as $2(2A)$. Applying the sine double angle formula:

$$\begin{aligned}\sin(4A) &= \sin(2\cdot 2A)\\ &= 2\sin(2A)\cos(2A)\end{aligned}$$

Substitute $\sin(2A) = 2\sin A\cos A$ and $\cos(2A) = \cos^2 A - \sin^2 A$:

$$\begin{aligned}\sin(4A) &= 2(2\sin A\cos A)(\cos^2 A - \sin^2 A)\\ &= 4\sin A\cos A(\cos^2 A - \sin^2 A)\\ &= 4\sin A\cos^3 A - 4\sin^3 A\cos A\end{aligned}$$

5. Compute the following limits (you may *not* use L'Hôpital's rule here):

(a) $\lim\limits_{x\to 0}\dfrac{1-\cos x}{x}$ (Hint: Multiply the numerator and denominator by $1+\cos x$.)

$$\lim_{x\to 0}\frac{1-\cos x}{x}\cdot\frac{1+\cos x}{1+\cos x} = \lim_{x\to 0}\frac{1-\cos^2 x}{x(1+\cos x)} = \lim_{x\to 0}\frac{\sin^2 x}{x(1+\cos x)}$$
$$= \lim_{x\to 0}\frac{\sin x}{x}\cdot\lim_{x\to 0}\frac{\sin x}{1+\cos x} = 1\cdot\frac{0}{1+1} = 1\cdot 0 = 0.$$

(b) $\lim_{x\to 0} \dfrac{\tan x}{x}$ (Hint: Rewrite $\tan x$ in terms of $\sin x$ and $\cos x$.)

$\lim_{x\to 0} \dfrac{\sin x/\cos x}{x} = \lim_{x\to 0} \dfrac{\sin x}{x} \cdot \dfrac{1}{\cos x} = \lim_{x\to 0} \dfrac{\sin x}{x} \cdot \lim_{x\to 0} \dfrac{1}{\cos x} = 1 \cdot \dfrac{1}{1} = 1.$

(c) $\lim_{x\to 0} \dfrac{\sin(2x)}{x\cos(2x)}$ (Hint: Use $\sin(2x) = 2\sin x \cos x$.)

$\lim_{x\to 0} \dfrac{2\sin x\cos x}{x\cos(2x)} = 2\lim_{x\to 0} \dfrac{\sin x}{x} \cdot \lim_{x\to 0} \dfrac{\cos x}{\cos(2x)} = 2(1) \cdot \dfrac{1}{1} = 2.$

(d) $\lim_{x\to\infty} \dfrac{\cos^2 x}{x^2}$ (Hint: What are the lower and upper bounds for the numerator?)

Since $0 \le \cos^2 x \le 1$ for all x, and the denominator gets bigger and bigger, this goes to 0.

6. Prove the following identities. In each case, you must start from the expression on the left and arrive at the right. You may not manipulate both sides. Try rewriting each in terms of $\sin x$ and $\cos x$.

(a) $\dfrac{\sec x}{\csc x} = \tan x$

$\dfrac{\sec x}{\csc x} = \dfrac{\frac{1}{\cos x}}{\frac{1}{\sin x}} = \dfrac{1}{\cos x} \cdot \dfrac{\sin x}{1} = \dfrac{\sin x}{\cos x} = \tan x.$

(b) $\sec\theta - \cos\theta = \sin\theta\tan\theta$

$\sec\theta - \cos\theta = \dfrac{1}{\cos\theta} - \cos\theta = \dfrac{1}{\cos\theta} - \dfrac{\cos^2\theta}{\cos\theta} = \dfrac{1-\cos^2\theta}{\cos\theta} = \dfrac{\sin^2\theta}{\cos\theta} = \sin\theta \cdot \dfrac{\sin\theta}{\cos\theta} = \sin\theta\tan\theta.$

(c) $\cos x(\sec x - \cos x) = \sin^2 x$

$\cos x(\sec x - \cos x) = \cos x \sec x - \cos^2 x = \cos x\left(\frac{1}{\cos x}\right) - \cos^2 x = 1 - \cos^2 x = \sin^2 x.$

(d) $\tan^2 x\cos^2 x + \cos^2 x = 1$

$\tan^2 x\cos^2 x + \cos^2 x = \left(\frac{\sin^2 x}{\cos^2 x}\right)\cos^2 x + \cos^2 x = \sin^2 x + \cos^2 x = 1.$

7. Prove the following identities. In each case, you must start from the expression on the left and arrive at the right. You may not manipulate both sides.

You will need the identities:

$$\sin(2A) = 2\sin A\cos B$$
$$\cos(2A) = \cos^2 A - \sin^2 A = 1 - 2\sin^2 A = 2\cos^2 A - 1$$

(a) $\dfrac{\sin(2x)}{\sin x} - \dfrac{\cos(2x)}{\cos x} = \sec x$

$\dfrac{2\sin x\cos x}{\sin x} - \dfrac{\cos(2x)}{\cos x} = 2\cos x - \dfrac{2\cos^2 x - 1}{\cos x}$

$= \dfrac{2\cos^2 x}{\cos x} - \dfrac{2\cos^2 x - 1}{\cos x} = \dfrac{2\cos^2 x - 2\cos^2 x + 1}{\cos x} = \dfrac{1}{\cos x} = \sec x.$

(b) $(\sin x + \cos x)^2 = 1 + \sin(2x)$

$(\sin x + \cos x)^2 = \sin^2 x + 2\sin x\cos x + \cos^2 x$

$= (\sin^2 x + \cos^2 x) + 2\sin x\cos x = 1 + \sin(2x).$

(c) $\dfrac{1 - \cos(2x)}{\sin(2x)} = \tan x$

$\dfrac{1 - (1 - 2\sin^2 x)}{2\sin x\cos x} = \dfrac{2\sin^2 x}{2\sin x\cos x} = \dfrac{\sin x}{\cos x} = \tan x.$

(d) $\cos^4 x - \sin^4 x = \cos(2x)$ (Hint: This is a difference of squares. Challenge question! Not on the test, but worth a try if you've done all the rest.)

$\cos^4 x - \sin^4 x = (\cos^2 x - \sin^2 x)(\cos^2 x + \sin^2 x)$
$= (\cos^2 x - \sin^2 x)(1) = \cos(2x).$

Differentiating Trig Functions

1. What function is closely approximated by $\frac{\tan(x+0.00001)-\tan x}{0.00001}$? Explain.

   This expression should closely approximate $\sec^2 x$, since

   $$\sec^2 x = \frac{d}{dx}\tan x = \lim_{h\to 0}\frac{\tan(x+h)-\tan x}{h} \approx \frac{\tan(x+0.00001)-\tan x}{0.00001}.$$

2. Find the equation of the tangent line for each of the following functions at the given point.

   (a) $f(x) = \sin x$ at $x = \pi$

   First, find the y-coordinate: $f(\pi) = \sin(\pi) = 0$.
   Find the derivative: $f'(x) = \cos x$.
   The slope at $x = \pi$ is $m = f'(\pi) = \cos(\pi) = -1$.
   Using point-slope form: $y - 0 = -1(x - \pi) \implies y = -x + \pi$.

   (b) $f(x) = \tan x$ at $x = \frac{\pi}{4}$

   y-coordinate: $f(\frac{\pi}{4}) = \tan(\frac{\pi}{4}) = 1$.
   Derivative: $f'(x) = \sec^2 x$.
   Slope: $m = \sec^2(\frac{\pi}{4}) = (\sqrt{2})^2 = 2$.
   Equation: $y - 1 = 2(x - \frac{\pi}{4}) \implies y = 2x - \frac{\pi}{2} + 1$.

   (c) $f(x) = \cos(2x)$ at $x = \frac{\pi}{6}$

   y-coordinate: $f(\frac{\pi}{6}) = \cos(\frac{\pi}{3}) = \frac{1}{2}$.
   Derivative (using chain rule): $f'(x) = -2\sin(2x)$.
   Slope: $m = -2\sin(\frac{\pi}{3}) = -2(\frac{\sqrt{3}}{2}) = -\sqrt{3}$.
   Equation: $y - \frac{1}{2} = -\sqrt{3}(x - \frac{\pi}{6})$.

   (d) $f(x) = x\sin x$ at $x = \frac{\pi}{2}$

   y-coordinate: $f(\frac{\pi}{2}) = \frac{\pi}{2}\sin(\frac{\pi}{2}) = \frac{\pi}{2}(1) = \frac{\pi}{2}$.
   Derivative (using product rule): $f'(x) = (1)\sin x + x(\cos x)$.
   Slope: $m = \sin(\frac{\pi}{2}) + \frac{\pi}{2}\cos(\frac{\pi}{2}) = 1 + \frac{\pi}{2}(0) = 1$.
   Equation: $y - \frac{\pi}{2} = 1(x - \frac{\pi}{2}) \implies y = x$.

   (e) $f(x) = x^2\cos x$ at $x = \pi$

   y-coordinate: $f(\pi) = \pi^2\cos(\pi) = \pi^2(-1) = -\pi^2$.
   Derivative (using product rule): $f'(x) = 2x\cos x + x^2(-\sin x)$.
   Slope: $m = 2(\pi)\cos(\pi) - \pi^2\sin(\pi) = 2\pi(-1) - \pi^2(0) = -2\pi$.
   Equation: $y - (-\pi^2) = -2\pi(x - \pi) \implies y + \pi^2 = -2\pi x + 2\pi^2 \implies y = -2\pi x + \pi^2$.

3. For each of the following functions, find the linear approximation $L(x)$ at the indicated point, then use that approximation to estimate the given value. Provide exact values using fractions, π, and roots.

   (a) $f(x) = \sin(2x)$ at $x = \frac{\pi}{3}$. Estimate $\sin(\frac{7\pi}{18})$.

   $f'(x) = 2\cos(2x) \implies f'(\frac{\pi}{3}) = 2\cos(\frac{2\pi}{3}) = 2(-\frac{1}{2}) = -1$.
   $f(\frac{\pi}{3}) = \sin(\frac{2\pi}{3}) = \frac{\sqrt{3}}{2}$.
   $L(x) = \frac{\sqrt{3}}{2} - 1(x - \frac{\pi}{3})$.
   Estimation: $\sin(2\cdot\frac{7\pi}{18}) \approx L(\frac{7\pi}{18}) = \frac{\sqrt{3}}{2} - (\frac{7\pi}{18} - \frac{6\pi}{18}) = \frac{\sqrt{3}}{2} - \frac{\pi}{18}$.

(b) $h(x) = \tan x$ at $x = \frac{\pi}{4}$. Estimate $\tan(\frac{5\pi}{24})$.

$h'(x) = \sec^2 x \implies h'(\frac{\pi}{4}) = (\sqrt{2})^2 = 2.$
$h(\frac{\pi}{4}) = 1.$
$L(x) = 1 + 2(x - \frac{\pi}{4}).$
Estimation: $\tan\frac{5\pi}{24} \approx L(\frac{5\pi}{24}) = 1 + 2(\frac{5\pi}{24} - \frac{6\pi}{24}) = 1 - \frac{\pi}{12}.$

(c) $f(x) = \cos^2 x$ at $x = \frac{\pi}{6}$. Estimate $\cos^2(\frac{\pi}{5})$.

$f'(x) = -2\cos x \sin x = -\sin(2x) \implies f'(\frac{\pi}{6}) = -\sin(\frac{\pi}{3}) = -\frac{\sqrt{3}}{2}.$
$f(\frac{\pi}{6}) = (\frac{\sqrt{3}}{2})^2 = \frac{3}{4}.$
$L(x) = \frac{3}{4} - \frac{\sqrt{3}}{2}(x - \frac{\pi}{6}).$
Estimation: $\cos^2(\frac{\pi}{5}) \approx L(\frac{\pi}{5}) = \frac{3}{4} - \frac{\sqrt{3}}{2}(\frac{6\pi}{30} - \frac{5\pi}{30}) = \frac{3}{4} - \frac{\sqrt{3}\pi}{60}.$

(d) $g(x) = \csc x$ at $x = \frac{\pi}{2}$. Estimate $\csc(\frac{11\pi}{20})$.

$g'(x) = -\csc x \cot x \implies g'(\frac{\pi}{2}) = -(1)(0) = 0.$
$g(\frac{\pi}{2}) = 1.$
$L(x) = 1 + 0(x - \frac{\pi}{2}) = 1.$
Estimation: $\csc(\frac{11\pi}{20}) \approx L(\frac{11\pi}{20}) = 1.$

(e) $f(x) = \cos(x + \frac{\pi}{4})$ at $x = 0$. Estimate $\cos(\frac{\pi}{4} + \frac{1}{10})$.

$f'(x) = -\sin(x + \frac{\pi}{4}) \implies f'(0) = -\frac{\sqrt{2}}{2}.$
$f(0) = \frac{\sqrt{2}}{2}.$
$L(x) = \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}x.$
Estimation: $x = 0.1 \implies \cos(\frac{\pi}{4} + \frac{1}{10}) \approx L(\frac{1}{10}) = \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{20} = \frac{9\sqrt{2}}{20}.$

4. For each function, find all critical points in the interval $[0, 2\pi]$. After differentiating, you will need the double angle formulas:

$$\sin(2A) = 2\sin A\cos B$$
$$\cos(2A) = \cos^2 A - \sin^2 A = 1 - 2\sin^2 A = 2\cos^2 A - 1$$

You will also need to factor or use the quadratic formula.

(a) $y = 2\sin x + \cos(2x)$

$y' = 2\cos x - 2\sin(2x) = 2\cos x - 4\sin x\cos x.$
Set $y' = 0$: $2\cos x(1 - 2\sin x) = 0.$
$\cos x = 0 \implies x = \frac{\pi}{2}, \frac{3\pi}{2}.$
$1 - 2\sin x = 0 \implies \sin x = \frac{1}{2} \implies x = \frac{\pi}{6}, \frac{5\pi}{6}.$
Critical Points: $x = \{\frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}\}.$

(b) $y = \cos(2x) - 2\cos x$

$y' = -2\sin(2x) + 2\sin x = -4\sin x\cos x + 2\sin x.$
Set $y' = 0$: $-2\sin x(2\cos x - 1) = 0.$
$-2\sin x = 0 \implies x = 0, \pi, 2\pi.$
$2\cos x - 1 = 0 \implies \cos x = \frac{1}{2} \implies x = \frac{\pi}{3}, \frac{5\pi}{3}.$
Critical Points: $x = \{0, \frac{\pi}{3}, \pi, \frac{5\pi}{3}, 2\pi\}.$

(c) $y = \cos(2x) + 2\sin x$

$y' = -2\sin(2x) + 2\cos x = -4\sin x\cos x + 2\cos x.$
Set $y' = 0$: $2\cos x(1 - 2\sin x) = 0.$

$2\cos x = 0 \implies x = \frac{\pi}{2}, \frac{3\pi}{2}$.
$1 - 2\sin x = 0 \implies \sin x = \frac{1}{2} \implies x = \frac{\pi}{6}, \frac{5\pi}{6}$.
Critical Points: $x = \{\frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}\}$.

(d) $y = \sin x + \frac{1}{2}\cos(2x)$

$y' = \cos x - \sin(2x) = \cos x - 2\sin x \cos x = \cos x(1 - 2\sin x)$.
Set $y' = 0$: $\cos x = 0$ or $1 - 2\sin x = 0$.
$\cos x = 0 \implies x = \frac{\pi}{2}, \frac{3\pi}{2}$.
$\sin x = \frac{1}{2} \implies x = \frac{\pi}{6}, \frac{5\pi}{6}$.
Critical Points: $x = \{\frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}\}$.

(e) $y = \sin(2x) - 3\sin x$ (Note: This one requires inverse trig functions.)

$y' = 2\cos(2x) - 3\cos x = 2(2\cos^2 x - 1) - 3\cos x = 4\cos^2 x - 3\cos x - 2$.
Set $y' = 0$ and use the quadratic formula for $\cos x$:

$$\cos x = \frac{3 \pm \sqrt{(-3)^2 - 4(4)(-2)}}{2(4)} = \frac{3 \pm \sqrt{41}}{8}$$

Approximations:

i. $\cos x = \frac{3+6.403}{8} \approx 1.25$

ii. $\cos x = \frac{3-6.403}{8} \approx -0.425$

Rejection: We reject $\cos x \approx 1.25$ because the maximum value of $\cos x$ is 1. No real value of x satisfies this.
Using $\cos x \approx -0.425$, the valid critical points are:

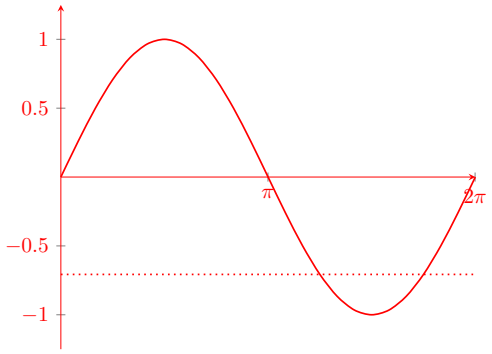
$x \approx \arccos(-0.425) \approx 2.01$ and $x \approx 2\pi - \arccos(-0.425) \approx 4.27$.

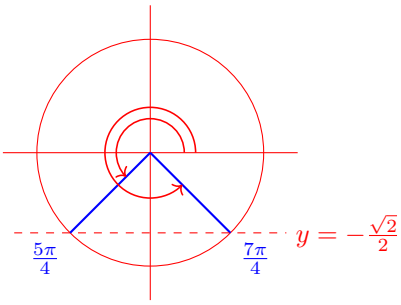
Inverse Trig Functions

For Q1-5, solve the given equation on the given domain. In each case, draw the graph of the respective base function (e.g. $\cos x$ for Q3), and indicate solutions on it using horizontal lines. Also draw the unit circles for each, showing all solutions.

1. $2\sin x + \sqrt{2} = 0$ for $0 \le x \le 2\pi$

   $2\sin x = -\sqrt{2} \implies \sin x = -\frac{\sqrt{2}}{2}$.

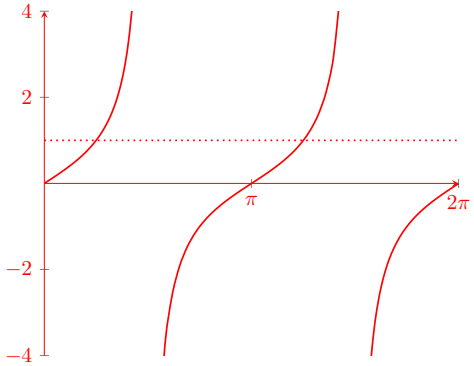


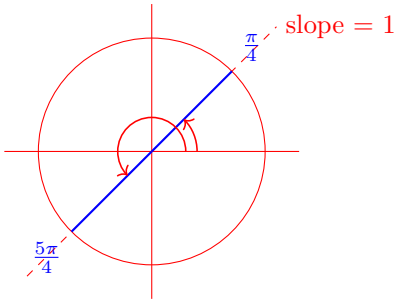


   There are two solutions: $x = \frac{5\pi}{4}$ and $x = \frac{7\pi}{4}$.

2. $3\tan x - 1 = 2$ for $0 \le x \le 2\pi$

   $3\tan x = 3 \implies \tan x = 1$.

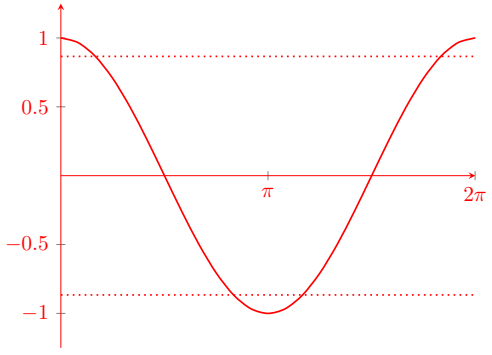


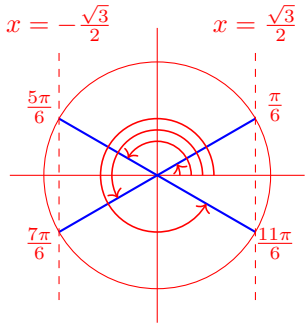


   There are two solutions: $x = \frac{\pi}{4}$ and $x = \frac{5\pi}{4}$.

3. $4\cos^2 x - 1 = 2$ for $0 \le x \le 2\pi$

   $4\cos^2 x = 3 \implies \cos^2 x = \frac{3}{4} \implies \cos x = \pm\frac{\sqrt{3}}{2}$.

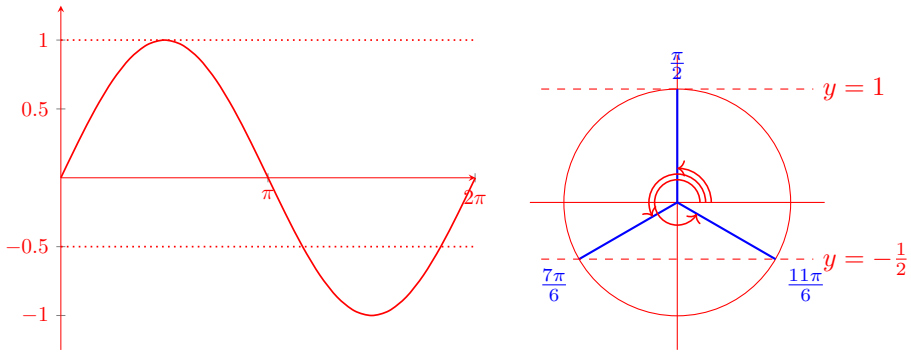




   The solutions are $x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}$, and $\frac{11\pi}{6}$.

4. $2\sin^2 x - \sin x - 1 = 0$ for $0 \le x \le 2\pi$

   Factoring the quadratic gives $(2\sin x + 1)(\sin x - 1) = 0$. This implies $\sin x = -\frac{1}{2}$ or $\sin x = 1$.



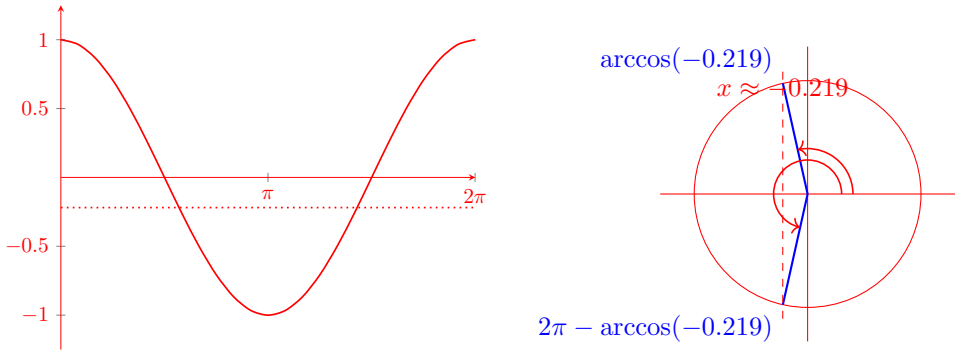
   The solutions are $x = \frac{\pi}{2}, \frac{7\pi}{6}$, and $\frac{11\pi}{6}$.

5. $2\cos^2 x + 5\cos x + 1 = 0$ for $0 \le x \le 2\pi$

   Using the quadratic formula: $\cos x = \frac{-5 \pm \sqrt{5^2 - 4(2)(1)}}{2(2)} = \frac{-5 \pm \sqrt{17}}{4}$.

   Approximating these values: $\cos x = \frac{-5-\sqrt{17}}{4} \approx -2.28$. This is rejected because the range of the cosine function is $[-1, 1]$, meaning $\cos x$ cannot be less than -1.

   $\cos x = \frac{-5+\sqrt{17}}{4} \approx -0.219$. Using the inverse cosine function: $x = \arccos\left(\frac{-5+\sqrt{17}}{4}\right)$.



   The approximate solutions are $x \approx 1.794$ rad and $x \approx 4.489$ rad.

6. Explain why $f(x) = \sin x$ is considered invertible on the domain $[-\frac{\pi}{2}, \frac{\pi}{2}]$ but not on the domain $[-2\pi, 2\pi]$.

   On the domain $[-2\pi, 2\pi]$, the sine function is periodic and fails the horizontal line test because multiple inputs yield the same output. By restricting the domain to $[-\frac{\pi}{2}, \frac{\pi}{2}]$, the function becomes strictly increasing (one-to-one), which allows for the definition of a unique inverse function, $\arcsin x$.

7. Describe the geometric relationship between the graph of a trigonometric function (on its restricted domain) and the graph of its inverse.

   The graphs of a trigonometric function and its inverse are reflections of each other across the identity line $y = x$. This reflection swaps the x and y coordinates of every point on the graph.

8. For what specific interval of x is the statement $\cos^{-1}(\cos x) = x$ true? Why is it not true for all real numbers?

   The statement is true only for $x \in [0, \pi]$. It is not true for all real numbers because the range of the inverse cosine function is restricted to $[0, \pi]$. For any x outside this interval, $\cos^{-1}(\cos x)$ will return the "principal" angle within the restricted range that has the same sine value.

9. Complete the following conceptual mappings:

   - $\sin^{-1}(x)$ takes a _____ and returns an _____.
   - $\tan^{-1}(x)$ takes a _____ and returns an _____.

   - $\sin^{-1}(x)$ takes a **ratio (y-coordinate on unit circle)** and returns an **angle (in radians)**.
   - $\tan^{-1}(x)$ takes a **ratio (slope)** and returns an **angle (in radians)**.

10. Describe what happens to the vertical asymptotes of $y = \tan x$ when the function is inverted to $y = \arctan x$. What is $\lim_{x \to -\infty} \arctan(x)$?

    The vertical asymptotes of $y = \tan x$ located at $x = -\frac{\pi}{2}$ and $x = \frac{\pi}{2}$ become horizontal asymptotes for the inverse function $y = \arctan x$ at $y = -\frac{\pi}{2}$ and $y = \frac{\pi}{2}$.

    As $x \to -\infty$, $\arctan(x)$ approaches its lower HA of $\frac{\pi}{2}$.

Differentiating Inverse Trig Functions

1. To invert $\cot x$, we restrict it to the domain $0 < x < \pi$. We call its inverse arccot x.

   Find the derivative of $y = \text{arccot}(x)$ using the derivative of $\cot(x)$ and implicit differentiation. You will need to find an identity relating $\csc^2 y$ to $\cot^2 y$ along the way.

   Let $y = \text{arccot}(x)$. By the definition of inverse functions, this is equivalent to:

   $$x = \cot(y)$$

   where $y \in (0, \pi)$.

   Differentiate both sides with respect to x using implicit differentiation:

   $$\frac{d}{dx}(x) = \frac{d}{dx}(\cot(y))$$

   $$1 = -\csc^2(y) \cdot \frac{dy}{dx}$$

   Now, isolate $\frac{dy}{dx}$:

   $$\frac{dy}{dx} = \frac{1}{-\csc^2(y)} = -\frac{1}{\csc^2(y)}$$

   We need to express $\csc^2(y)$ in terms of x. We divide the Pythagorean identity $\cos^2 y + \sin^2 y = 1$ by $\sin^2 y$ to get $\frac{\cos^2 y}{\sin^2 y} + \frac{\sin^2 y}{\sin^2 y} = \frac{1}{\sin^2 y}$

   That is:

   $$\csc^2(y) = 1 + \cot^2(y)$$

   Since we know $x = \cot(y)$, we can substitute:

   $$\csc^2(y) = 1 + x^2$$

   Substitute this back into our expression for the derivative:

   $$\frac{dy}{dx} = -\frac{1}{1 + x^2}$$

   Final Result:

   $$\frac{d}{dx}(\text{arccot}(x)) = -\frac{1}{1 + x^2}$$

2. Find the derivative of $f(x) = x^2 \arctan(x)$.

   We use the **product rule**: $f'(x) = \frac{d}{dx}[x^2] \cdot \arctan(x) + x^2 \cdot \frac{d}{dx}[\arctan(x)]$.

   $$f'(x) = 2x \arctan(x) + x^2 \left(\frac{1}{1 + x^2}\right)$$

   $$f'(x) = 2x \arctan(x) + \frac{x^2}{1 + x^2}$$

3. Find the derivative of $g(x) = \arcsin(e^x)$.

We use the **chain rule**: Let $u = e^x$. Then $\frac{du}{dx} = e^x$.

$$\frac{d}{dx}[\arcsin(u)] = \frac{1}{\sqrt{1-u^2}} \cdot \frac{du}{dx}$$
$$g'(x) = \frac{1}{\sqrt{1-(e^x)^2}} \cdot e^x$$
$$g'(x) = \frac{e^x}{\sqrt{1-e^{2x}}}$$

4. Find the derivative of $h(x) = \frac{\arccos(x)}{x+1}$.

We use the **quotient rule**: $\frac{f'g-fg'}{g^2}$.

$$h'(x) = \frac{\left(-\frac{1}{\sqrt{1-x^2}}\right)(x+1) - \arccos(x)(1)}{(x+1)^2}$$
$$h'(x) = \frac{-\frac{x+1}{\sqrt{1-x^2}} - \arccos(x)}{(x+1)^2}$$

5. Find the derivative of $y = \arctan(\sqrt{x})$.

We use the **chain rule**: Let $u = \sqrt{x} = x^{1/2}$. Then $\frac{du}{dx} = \frac{1}{2\sqrt{x}}$.

$$\frac{dy}{dx} = \frac{1}{1+u^2} \cdot \frac{du}{dx}$$
$$\frac{dy}{dx} = \frac{1}{1+(\sqrt{x})^2} \cdot \frac{1}{2\sqrt{x}}$$
$$\frac{dy}{dx} = \frac{1}{2\sqrt{x}(1+x)}$$

6. Find the derivative of $f(x) = \ln(\arcsin(x))$.

We use the **chain rule** for the natural logarithm and the inverse sine function.

$$f'(x) = \frac{1}{\arcsin(x)} \cdot \frac{d}{dx}[\arcsin(x)]$$
$$f'(x) = \frac{1}{\arcsin(x)} \cdot \frac{1}{\sqrt{1-x^2}}$$
$$f'(x) = \frac{1}{\arcsin(x)\sqrt{1-x^2}}$$

7. Use linear approximation to estimate the value of $\arcsin(0.51)$.

Let $f(x) = \arcsin(x)$. We choose the nearby point $a = 0.5$ (or $1/2$) because we know the exact value of $\arcsin(0.5)$.

1. Evaluate the function at a:

$$f(0.5) = \arcsin(0.5) = \frac{\pi}{6}$$

2. Find the derivative and evaluate at a:

$$f'(x) = \frac{1}{\sqrt{1-x^2}}$$

$$f'(0.5) = \frac{1}{\sqrt{1-(0.5)^2}} = \frac{1}{\sqrt{0.75}} = \frac{2}{\sqrt{3}}$$

3. Build the linear approximation formula:

$$L(x) = f(a) + f'(a)(x-a)$$

$$L(x) = \frac{\pi}{6} + \frac{2}{\sqrt{3}}(x-0.5)$$

4. Estimate for $x = 0.51$:

$$L(0.51) = \frac{\pi}{6} + \frac{2}{\sqrt{3}}(0.51-0.5) = \frac{\pi}{6} + \frac{2}{\sqrt{3}}(0.01)$$

$$L(0.51) \approx 0.5236 + 0.0115 = 0.5351 \text{ radians.}$$

8. Use the tangent line to $f(x) = \arctan(x)$ at $x = 1$ to approximate $\arctan(1.05)$.

Let $f(x) = \arctan(x)$ and $a = 1$.

1. Evaluate the function at a:

$$f(1) = \arctan(1) = \frac{\pi}{4}$$

2. Find the derivative and evaluate at a:

$$f'(x) = \frac{1}{1+x^2}$$

$$f'(1) = \frac{1}{1+1^2} = \frac{1}{2}$$

3. Build the linear approximation formula:

$$L(x) = \frac{\pi}{4} + \frac{1}{2}(x-1)$$

4. Estimate for $x = 1.05$:

$$L(1.05) = \frac{\pi}{4} + \frac{1}{2}(1.05-1) = \frac{\pi}{4} + 0.025$$

$$L(1.05) \approx 0.7854 + 0.025 = 0.8104 \text{ radians.}$$

9. Use linear approximation to estimate the value of $f(0.1)$ for the composite function $f(x) = \arctan(1-2x)$.

Let $f(x) = \arctan(1-2x)$. We want to approximate $f(0.1)$, so we choose the center $a = 0$ because $f(0) = \arctan(1)$ is a known value.

1. Evaluate the function at $a = 0$:

$$f(0) = \arctan(1-2(0)) = \arctan(1) = \frac{\pi}{4}$$

2. Find the derivative using the chain rule: Let $u = 1-2x$. Then $\frac{du}{dx} = -2$.

$$f'(x) = \frac{1}{1+(1-2x)^2} \cdot (-2) = \frac{-2}{1+(1-2x)^2}$$

3. Evaluate the derivative at $a = 0$:

$$f'(0) = \frac{-2}{1+(1-0)^2} = \frac{-2}{1+1} = \frac{-2}{2} = -1$$

4. Build the linear approximation formula:

$$L(x) = f(0) + f'(0)(x-0)$$

$$L(x) = \frac{\pi}{4} - 1(x)$$

5. Estimate for $x = 0.1$:

$$L(0.1) = \frac{\pi}{4} - 0.1$$

$$L(0.1) \approx 0.7854 - 0.1 = 0.6854 \text{ radians.}$$

10. (a) What is $\tan(\arctan x)$?

$\tan(\arctan x) = x$

(b) Given the above, what should $\frac{d}{dx}\tan(\arctan x)$ be?

$\frac{d}{dx}\tan(\arctan x) = \frac{d}{dx}x = 1$

(c) Use the chain rule instead to differentiate $\tan(\arctan x)$.

$\frac{d}{dx}\tan(\arctan x) = \sec^2(\arctan x) \cdot \frac{1}{1+x^2}$.

(d) What must be true about your last two answers?

$1 = \sec^2(\arctan x) \cdot \frac{1}{1+x^2}$

(e) Let $y = \arctan x$. What kind of object is y? Label it on the triangle below.

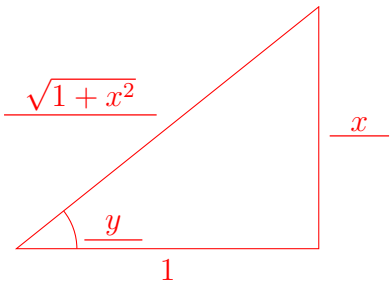
(f) If the bottom length of the triangle has length 1, label x on the triangle as well.

(g) Compute the length of the third side. Then, use the triangle to find $\sec(\arctan x)$. How does this fit in with your answer to (d)?

$$\frac{d}{dx}\arctan x = \frac{1}{1+x^2}$$

x^2.

This means that the denominator in (d) indeed cancels, as expected.

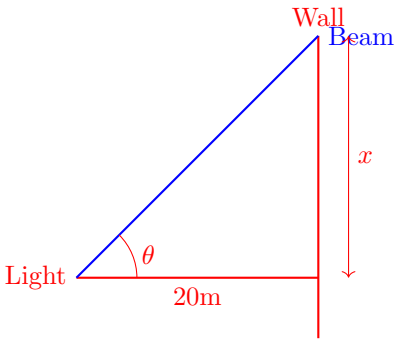


The hypotenuse is $\sqrt{1^2+x^2} = \sqrt{1+x^2}$.
From the triangle, $\sec(y) = \frac{\text{hyp}}{\text{adj}} = \frac{\sqrt{1+x^2}}{1}$.
Therefore, $\sec^2(\arctan x) = (\sqrt{1+x^2})^2 = 1 +$

Trig Related Rates

1. **The Rotating Searchlight**
   A searchlight 20 meters from a straight wall rotates at a constant rate of 0.5 rad/s. At what rate is the beam moving along the wall when the beam makes an angle of $\pi/4$ with the line perpendicular to the wall?

   Drawing Instructions: Draw a horizontal line of length 20 for the distance to the wall. At the end, draw a vertical line for the wall. Draw a diagonal beam from the light source to a point x on the wall. Label the angle between the horizontal and the beam as θ.



   Let x be the distance along the wall and θ be the angle. We have $x = 20\tan\theta$. Differentiating with respect to t:

$$\frac{dx}{dt} = 20\sec^2\theta\frac{d\theta}{dt}$$

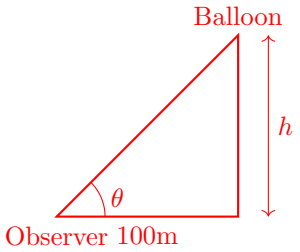
   Given $\frac{d\theta}{dt} = 0.5$ and $\theta = \pi/4$:

$$\sec^2(\pi/4) = (\sqrt{2})^2 = 2$$

$$\frac{dx}{dt} = 20(2)(0.5) = 20 \text{ m/s}$$

2. **The Rising Balloon**
   A hot air balloon rises vertically from a level field at a rate of 3 m/s. An observer is 100 meters from the lift-off point. At what rate is the angle of elevation changing when the balloon is 100 meters high?

   Drawing Instructions: Draw a right triangle with a horizontal base of 100 meters. The vertical side is the height h. Label the angle at the observer as θ.

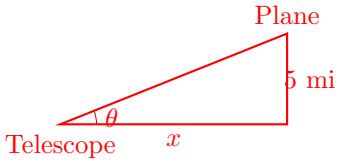


   $\tan\theta = \frac{h}{100} \implies 100\tan\theta = h$. Differentiating: $100\sec^2\theta\frac{d\theta}{dt} = \frac{dh}{dt}$. When $h = 100$, $\tan\theta = 1 \implies \theta = \pi/4$ and $\sec^2\theta = 2$.

$$100(2)\frac{d\theta}{dt} = 3 \implies \frac{d\theta}{dt} = \frac{3}{200} = 0.015 \text{ rad/s}$$

3. **Telescope Tracking**
A plane flies horizontally at 600 mph at an altitude of 5 miles. Find the rate of change of the telescope angle from the ground, θ, when the horizontal distance from the telescope is 12 miles.

Drawing Instructions: Draw a right triangle with vertical side 5 and horizontal side $x = 12$. Label the angle at the telescope as θ.



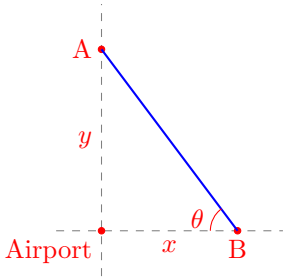
$\tan\theta = \frac{5}{x} \implies \theta = \arctan(5/x)$. Alternatively, $x = 5\cot\theta \implies \frac{dx}{dt} = -5\csc^2\theta\frac{d\theta}{dt}$. When $x = 12$, the hypotenuse is 13. Thus $\csc\theta = 13/5$. $600 = -5(169/25)\frac{d\theta}{dt} \implies 600 = -\frac{169}{5}\frac{d\theta}{dt}$.

$$\frac{d\theta}{dt} = -\frac{3000}{169} \text{ rad/h}$$

4. **Two Planes Crossing**
Plane A flies north at 400 km/h toward an airport. Plane B flies east at 300 km/h toward the same airport. Let θ be the bearing of A from B. Find $d\theta/dt$ when A is 4 km and B is 3 km from the airport.

Hint: you have three variables here – an angle and two distances. You'll need the quotient rule along the way.

Drawing Instructions: Draw a coordinate axes with the airport at $(0, 0)$. Place B at $(3, 0)$ and A at $(0, 4)$. Connect B and A, and label the angle at B as θ.

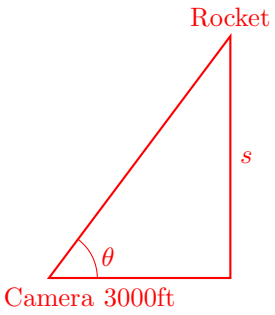


$\tan\theta = \frac{y}{x}$. Differentiating: $\sec^2\theta\frac{d\theta}{dt} = \frac{x\frac{dy}{dt} - y\frac{dx}{dt}}{x^2}$. At $x = 3, y = 4$, $\sec^2\theta = 25/9$. $\frac{dy}{dt} = -400$ and $\frac{dx}{dt} = -300$.

$$\frac{25}{9}\frac{d\theta}{dt} = \frac{3(-400) - 4(-300)}{9} = 0 \implies \frac{d\theta}{dt} = 0 \text{ rad/h}$$

5. **Rocket Tracking**
A camera 3000 ft from a launch pad tracks a rocket with height $s = 50t^2$. Find the rate of change of the angle θ when $t = 10$ s.

Drawing Instructions: Draw a right triangle with base 3000 and height s. Label the angle at the camera as θ.

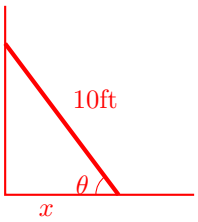


$\tan\theta = \frac{s}{3000} \implies \sec^2\theta\frac{d\theta}{dt} = \frac{1}{3000}\frac{ds}{dt}$. At $t = 10, s = 5000, \frac{ds}{dt} = 100t = 1000$. $\sec^2\theta = 1 + (5000/3000)^2 = 34/9$.

$$\frac{34}{9}\frac{d\theta}{dt} = \frac{1000}{3000} \implies \frac{d\theta}{dt} = \frac{3}{34} \text{ rad/s}$$

6. **Sliding Ladder**
A 10-ft ladder leans against a wall. The bottom slides away at 2 ft/s. How fast is the angle between the ladder and the floor changing when the bottom is 6 ft from the wall?

Drawing Instructions: Draw a vertical wall and horizontal floor. Draw a diagonal line for the ladder (length 10). Label the floor distance x and the angle θ.

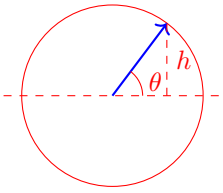


$x = 10\cos\theta \implies \frac{dx}{dt} = -10\sin\theta\frac{d\theta}{dt}$. When $x = 6, y = 8$, so $\sin\theta = 0.8$.

$$2 = -10(0.8)\frac{d\theta}{dt} \implies \frac{d\theta}{dt} = -0.25 \text{ rad/s}$$

7. **The Ferris Wheel**
A Ferris wheel (radius 10m) rotates at 1 rad/min. How fast is a passenger rising when they are 8m above the center's horizontal line?

Drawing Instructions: Draw a circle (radius 10) and its horizontal diameter. Draw a radius at angle θ and label the height h.



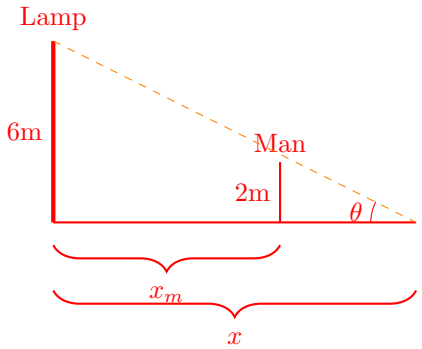
$h = 10\sin\theta \implies \frac{dh}{dt} = 10\cos\theta\frac{d\theta}{dt}$. When $h = 8, \sin\theta = 0.8 \implies \cos\theta = 0.6$.

$$\frac{dh}{dt} = 10(0.6)(1) = 6 \text{ m/min}$$

8. **The Moving Shadow**
A man 2 meters tall walks away from a 6-meter lamppost at 1.5 m/s. Let θ be the angle of the light beam hitting the ground. How fast is θ changing when the man is 8 meters from the post?

Hint: This one is harder, and involves similar triangles. Let the distance of the man from the post be x_m and the distance of the tip of the shadow from the post be x. You'll need to find a relationship between these two!

Drawing Instructions: Draw a large right triangle (height 6). Place a smaller vertical line (height 2) inside. Label the angle at the vertex on the ground as θ.



Let x_m be the distance of the man from the post, then $\frac{dx_m}{dt} = 1.5$ m/sec.

Let x be the total distance from the post to the tip of the shadow. $\cot\theta = \frac{x}{6}$.

Differentiating: $-\csc^2\theta\frac{d\theta}{dt} = \frac{1}{6}\frac{dx}{dt}$.

By similar triangles: $\frac{6}{2} = \frac{x}{x-x_m}$, or $3(x - x_m) = x$, which gives $2x = 3x_m$, or $x = 1.5x_m$.

When $x_m = 8$, $x = 1.5 \times 8 = 12$. The hypotenuse is $\sqrt{12^2 + 6^2} = 6\sqrt{5}$, so $\csc\theta = \sqrt{5}$.

We get $-5\frac{d\theta}{dt} = \frac{1}{6}\frac{dx}{dt} = \frac{1}{6} \cdot 1.5\frac{dx_m}{dt} = \frac{1}{6} \cdot 1.5 \cdot 1.5$

So

$$-5\frac{d\theta}{dt} = \frac{2.25}{6} \implies \frac{d\theta}{dt} = -0.075 \text{ rad/s}$$